Formula sheet - Probability & Statistics (PMTPS)

Law of total probability:

$$P(B) = P(B|A)P(A) + P(B|A')P(A')$$

Bayes' Rule:

$$P(A|B) = P(B|A) \cdot \frac{P(A)}{P(B)}$$

Expected value:

$$E(X) = \sum_x x \cdot P(X = x) \qquad \text{(discrete)}$$

$$E(X) = \int_x x \cdot f(x)\, dx \qquad \text{(continuous)}$$

Sample mean:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Variance:

$$\text{Var}(X) = \sum_x (x - E(X))^2 \cdot P(X = x) = E(X^2) - E(X)^2 \qquad \text{(discrete)}$$

$$\text{Var}(X) = \int_x (x - E(X))^2 \cdot f(x)\, dx = E(X^2) - E(X)^2 \qquad \text{(continuous)}$$

Sample variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$$

Covariance:

$$\text{Cov}(X, Y) = E\left((X - E(X)) \cdot (Y - E(Y))\right) = E(X \cdot Y) - E(X)E(Y)$$

Correlation coefficient:

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma(X) \cdot \sigma(Y)}$$

Variance of error terms:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \frac{\sum_{i=1}^{n} y_i^2 - \hat{\beta}_0 \cdot \sum_{i=1}^{n} y_i - \hat{\beta}_1 \cdot \sum_{i=1}^{n} x_i y_i}{n-2}$$

Coefficients of regression line:

Given a sample of n observations $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, the regression line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot x$ satisfies:

- $\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i y_i - n \cdot \overline{x} \cdot \overline{y}}{\sum_{i=1}^{n} x_i^2 - n \cdot \overline{x}^2} = \frac{\overline{xy} - \overline{x} \cdot \overline{y}}{\overline{x^2} - \overline{x}^2}$
- $\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \cdot \overline{x}$

Sample correlation coefficient:

$$r(x, y) = \frac{\text{Cov}(x, y)}{\sqrt{s_X^2 \cdot s_Y^2}} = \frac{\overline{xy} - \overline{x} \cdot \overline{y}}{\sqrt{(\overline{x^2} - \overline{x}^2) \cdot (\overline{y^2} - \overline{y}^2)}}$$

Likelihood function:

$$L(x_1, x_2, \ldots, x_n; \theta) = \prod_{i=1}^{n} f(x_i; \theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \ldots \cdot f(x_n; \theta)$$

Minimum sample size with given accuracy E:

$$n \geq \left(\frac{z_{\alpha/2} \cdot \sigma}{E} \right)^2$$

Minimum sample size for bounding type II error β

$$n \geq \frac{(z_{\alpha/2} + z_{\beta})^2 \cdot \sigma^2}{\delta^2}$$

Expected values and variance of common probability distributions:

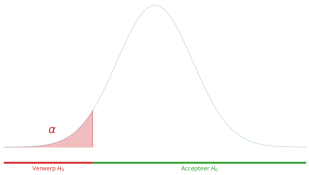
Distribution	PMF/PDF	CDF	Mean	Variance
		Discrete		
Uniform(a, b)	$P(X = k) = \frac{1}{b-a+1}$ $(k = a, a+1, \ldots, b)$	$F(k) = \begin{cases} 0 & x < a \\ \frac{k-a+1}{b-a+1} & a \leq k < b \\ 1 & k \geq b \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2-1}{12}$
Bernoulli(p)	$P(X = k) = \begin{cases} 1-p, & k = 0 \\ p, & k = 1 \end{cases}$	$F(k) = \begin{cases} 0 & k < 0 \\ 1-p & k = 0 \\ 1 & k \geq 1 \end{cases}$	p	$p(1-p)$
Binomial(n, p)	$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$	$F(k) = \sum_{i=0}^{k} \binom{n}{i} p^i (1-p)^{n-i}$	np	$np(1-p)$
Geometric(p)	$P(X = k) = (1-p)^{k-1} \cdot p$	$F(k) = \sum_{i=1}^{k} (1-p)^{k-1} \cdot p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	$P(X = k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$	$F(k) = \sum_{i=0}^{k} \frac{\lambda^i e^{-\lambda}}{i!}$	λ	λ
		Continuous		
Uniform(a, b)	$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise.} \end{cases}$	$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x < b \\ 1, & x \geq b \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal(μ, σ^2)	$f(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$\Phi\left(\frac{x-\mu}{\sigma}\right)$	μ	σ^2
Exponential(λ)	$f(x) = \lambda e^{-\lambda x}$, $x \geq 0$	$F(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Types of hypothesis tests

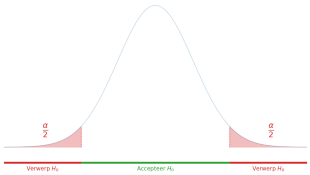
Test name	Test statistic	Probability distribution (under H_0)
Tests for the mean μ of a normal distribution		
σ known	$\overline{X}$	$N(\mu_0, \sigma/\sqrt{n})$
σ unknown	$T = \frac{\overline{X}-\mu_0}{S/\sqrt{n}}$	$t(\text{df} = n-1)$
Chi-square test (χ^2)		
Chi-square goodness-of-fit	$X^2 = \sum_i \frac{(O_i-E_i)^2}{E_i}$	χ^2 (df = #categories-1)

One-sided and two-sided hypothesis tests

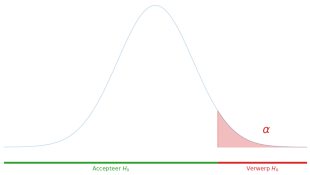
Left-sided
$H_0 : \mu \geq \mu_0$ vs. $H_1 : \mu < \mu_0$
Acceptance region: $[g, \infty)$
Critical region: $(-\infty, g)$



Two-sided
$H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$
Acceptance region: $[g_1, g_2]$
Critical region: $(-\infty, g_1)$ and (g_2, ∞)



Right-sided
$H_0 : \mu \leq \mu_0$ vs. $H_1 : \mu > \mu_0$
Acceptance region: $(-\infty, g]$
Critical region: (g, ∞)



Critical regions

Probability distribution (under H_0)	**Left-sided** $(-\infty, g)$	**Two-sided** $(-\infty, g_1)$ and (g_2, ∞)	**Right-sided** (g, ∞)
$N(\mu, \sigma)$	$g = \text{InvNorm}(\alpha, \mu, \sigma)$	$g_1 = \text{InvNorm}(\alpha/2, \mu, \sigma)$ $g_2 = \text{InvNorm}(1-\alpha/2, \mu, \sigma)$	$g = \text{InvNorm}(1-\alpha, \mu, \sigma)$
$t(\text{df})$	$g = \text{InvT}(\alpha, \text{df})$	$g_1 = \text{InvT}(\alpha/2, \text{df})$ $g_2 = \text{InvT}(1-\alpha/2, \text{df})$	$g = \text{InvT}(1-\alpha, \text{df})$
Boundary values that need to be solved with the "numeric solver":			
$\chi^2(\text{df})$	-	-	$\chi^2\text{cdf}(g, 10^{99}, \text{df}) = \alpha$

p-values (for given theoretical and observed test statistics T and t)

Probability distribution (under H_0)	**Left-sided test:** $P(T \leq t)$	**Right-sided test:** $P(T \geq t)$
$N(\mu, \sigma)$	$p = \text{normalcdf}(-10^{99}, t, \mu, \sigma)$	$p = \text{normalcdf}(t, 10^{99}, \mu, \sigma)$
$t(\text{df})$	$p = \text{tcdf}(-10^{99}, t, \text{df})$	$p = \text{tcdf}(t, 10^{99}, \text{df})$
$\chi^2(\text{df})$	-	$p = \chi^2\text{cdf}(t, 10^{99}, \text{df})$

Note: with two-sided tests, the p-value is equal to the minimum of $P(T \leq t)$ and $P(T \geq t)$.